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Inventor(s)

Soetaert; Michael et al.

FILTERED SPRAY HEAD ASSEMBLY

Abstract

A spray head assembly includes an inlet assembly configured to receive a fluid, a body fluidly coupled with the inlet assembly, a head portion fluidly removably coupled with the body, and a filter assembly. The head portion includes a plurality of nozzles configured to discharge fluid from the spray head assembly. The body and the head portion define an interior chamber. The filter assembly is configured to filter the fluid supplied from the inlet assembly to the head portion. The filter assembly is configured to be received within the interior chamber.

Inventors: Soetaert; Michael (Sheboygan, WI), Yee; Evan (Grafton, WI)

Applicant: Kohler Co. (Kohler, WI)

Family ID: 1000008453522

Assignee: Kohler Co. (Kohler, WI)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of and priority to U.S. Provisional Patent Application No. 63/557,040, filed Feb. 23, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] The present disclosure relates generally to water delivery devices, such as showerhead assemblies.

SUMMARY

[0003] One embodiment of the present disclosure relates to a showerhead assembly configured to filter a fluid (e.g., water) flowing therethrough.

[0004] One embodiment of the present disclosure relates to a spray head assembly. The spray head assembly includes an inlet assembly configured to receive a fluid, a body fluidly coupled with the inlet assembly, a head portion fluidly removably coupled with the body, and a filter assembly. The head portion includes a plurality of nozzles configured to discharge fluid from the spray head assembly. The body and the head portion define an interior chamber. The filter assembly is configured to filter the fluid supplied from the inlet assembly to the head portion. The filter assembly is configured to be received within the interior chamber. In some embodiments, separating the body and the head portion from each other facilitates providing access to the interior chamber to replace the filter assembly.

[0005] Another embodiment of the present disclosure relates to a spray head assembly. The spray head assembly includes an inlet assembly configured to receive a fluid, a body fluidly coupled with the inlet assembly, a head portion fluidly removably coupled with the body, the head portion including a plurality of nozzles configured to discharge the fluid from the spray head assembly, the body and the head portion defining an interior chamber, a filter housing defining a filter chamber and configured to be received within the interior chamber, and a filter assembly configured to filter the fluid supplied from the inlet assembly to the head portion, the filter assembly configured to be received within the filter chamber. The spray head assembly is configurable both as (i) a first spray head assembly including the body and the head portion defining a first shape configured to receive the filter housing and (ii) a second spray head assembly including the body and the head portion defining a second shape configured to receive the filter housing. The second shape is different than the first shape.

[0006] Still another embodiment of the present disclosure relates to a spray head assembly. The spray head assembly includes an inlet assembly configured to receive a fluid, a body fluidly coupled with the inlet assembly, the body including a connector flange, a head portion fluidly coupled with the body, the head portion including a plurality of nozzles configured to discharge the fluid from the spray head assembly, a filter housing defining a filter chamber, the filter housing including a first sidewall, a top wall configured to couple to the connector flange to couple the first sidewall with the body, and a second sidewall configured to couple to the head portion and overlap with at least a portion of the first sidewall, and a filter assembly configured to filter the fluid

supplied from the inlet assembly to the head portion, the filter assembly configured to be received within the filter chamber of the filter housing.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0007] The disclosure will become more fully understood from the following detailed description, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements, in which:

[0008] FIG. 1 is a bottom perspective view of a spray head assembly, according to an embodiment;

[0009] FIG. 2 is a top perspective view of the spray head assembly of FIG. 1;

[0010] FIG. 3 is a front view of the spray head assembly of FIG. 1;

[0011] FIG. 4 is a side view of the spray head assembly of FIG. 1;

[0012] FIG. 5 is a top view of the spray head assembly of FIG. 1;

[0013] FIG. 6 is a bottom view of the spray head assembly of FIG. 1;

[0014] FIG. 7 is a bottom perspective view of a spray head assembly, according to another embodiment;

[0015] FIG. 8 is a top perspective view of the spray head assembly of FIG. 7;

[0016] FIG. 9 is a front view of the spray head assembly of FIG. 7;

[0017] FIG. 10 is a side view of the spray head assembly of FIG. 7;

[0018] FIG. 11 is a top view of the spray head assembly of FIG. 7;

[0019] FIG. 12 is a bottom view of the spray head assembly of FIG. 7;

[0020] FIG. 13 is a side cross-sectional view of the spray head assembly of FIG. 1;

[0021] FIG. 14 is a side cross-sectional view of the spray head assembly of FIG. 7;

[0022] FIG. 15A is a bottom perspective view of the spray head assembly of FIG. 1 illustrating a first disassembly operation to access a filter element, according to an embodiment;

[0023] FIG. 15B is a bottom perspective view of the spray head assembly of FIG. 1 illustrating a second disassembly operation to access a filter assembly, according to an embodiment;

[0024] FIG. 16 is a bottom view of a spray head assembly, according to another embodiment;

[0025] FIG. 17 is a side view of the spray head assembly of FIG. 16;

[0026] FIG. 18 is a top view of the spray head assembly of FIG. 16;

[0027] FIG. 19A is a perspective view of a filter housing, according to an embodiment;

[0028] FIG. 19B is a side cross-sectional view of the filter housing of FIG. 19A;

[0029] FIG. 20 is an exploded bottom perspective view of a spray head assembly including a filter assembly, according to an embodiment;

[0030] FIG. 21 is an exploded top perspective view of the spray head assembly of FIG. 20;

[0031] FIG. 22 is an exploded bottom perspective view of a spray head assembly including a filter assembly, according to an embodiment;

[0032] FIG. 23 is a perspective view of a filter assembly, according to an embodiment; and

[0033] FIG. 24 is a perspective view of a spray head assembly including a filter, according to an embodiment.

DETAILED DESCRIPTION

[0034] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0035] Embodiments of the present disclosure relate to a spray head assembly, such as a showerhead assembly, that includes a removable filter assembly for removing contaminants and

impurities from a water supply. In some embodiments, the spray head assembly is a self-contained showerhead assembly that includes a single inlet connector and spray face that are integrally formed into a shared housing structure that also contains the removable filter assembly. The spray head assembly may also include a filter support structure that supports a common filter assembly design within various different spray head structures and/or aesthetics. The filter support structure also enables replacement of the filter assembly by hand without the use of mechanical fasteners. The spray head assembly includes an inlet assembly configured to receive a supply of fluid, a body fluidly coupled with the inlet assembly, and a head portion configured to receive the fluid from the body and discharge the fluid through a plurality of nozzles. The body defines an interior chamber to receive the filter support structure. The filter support structure houses the filter assembly. The body and the head portion may be removably coupled (e.g., rotatably removably coupled) to each other, which simplifies filter replacement operations, and can ensure a more watertight seal between the filter assembly and the spray head assembly. When the body and the head portion are separated, a bottom wall of the filter support may remain coupled with the head portion and a top wall of the filter support may remain coupled with the body, thereby facilitating access to the filter support structure to replace the filter assembly.

[0036] Referring to FIGS. **1-12**, a spray head assembly (e.g., a showerhead assembly, a faucet spray head assembly, etc.) is shown as spray head assembly **10**, according to an embodiment. The spray head assembly **10** includes an inlet assembly (e.g., an inlet, etc.), shown as inlet assembly **14**, configured to fluidly couple to a fluid source (e.g., a well, a plumbing system, etc.) to receive a supply of fluid (e.g., water); an outlet assembly (e.g., an outlet, etc.), shown as head portion **18**, configured to discharge the fluid; and a body assembly (e.g., a shell, an outer body, a casing, etc.), shown as body **22**, extending between the head portion **18** and the inlet assembly **14** and configured to fluidly connect the inlet assembly **14** with the head portion **18**. By way of example, the body **22** defines a channel, a passage, a bore, etc. to supply the fluid received by the inlet assembly **14** to the head portion **18** to be discharged therefrom. In at least one embodiment, the spray head assembly **10** is a showerhead assembly that is configured to receive unfiltered water from a water supply (e.g., a water supply conduit, a water supply pipe, a water supply line, etc.) and to distribute a spray of clean, filtered water into a shower enclosure. The spray head assembly **10** may be configured to discharge water in a spray pattern defined by the head portion **18**.

[0037] The inlet assembly **14** includes a fluid communication member (e.g., a tube, a pipe, a conduit, etc.), shown as connector **26**, configured to fluidly couple the spray head assembly **10** to a fluid delivery component (e.g., a pipe, a spout, a fixture, a shower arm, etc.) such that the spray head assembly **10** can receive the fluid from the fluid source. In some embodiments, the fluid delivery component is a water supply conduit that is configured to supply water at residential and/or commercial water supply pressures. In some embodiments, the connector **26** includes threads that screw to the fluid delivery component. In such embodiments, the connector **26** facilitates selectively coupling the spray head assembly **10** with the fluid delivery component. In some embodiments, the spray head assembly **10** is configured as a pull-out spray head, where the spray head assembly **10** can be selectively removed from and reattached to the fluid delivery component. Alternatively, the spray head assembly **10** may be fixedly coupled to the fluid delivery component.

[0038] In some embodiments, the inlet assembly **14** is configured to pivotably couple the spray head assembly **10** to the fluid delivery component so that the spray head assembly **10** can pivot relative to the inlet assembly **14**. The inlet assembly **14** includes a ball joint **30** (e.g., a spherical element, etc.) attached to the connector **26**. When the inlet assembly **14** fluidly couples the spray head assembly **10** to the fluid delivery component, the spray head assembly **10** is pivotable about the ball joint **30** (e.g., pivotable about a center point of the ball joint **30**) relative to the inlet assembly **14** and the fluid delivery component. The connector **26** and the ball joint **30** together define a fluid passage **34** that extends through the ball joint **30** and the connector **26** to supply the

fluid to the head portion **18** (e.g., through the body **22**) to be discharged therefrom. In some embodiments, the inlet assembly **14** includes a flow regulator **38** to regulate the pressure of the flow of the fluid to the body **22** and the head portion **18**. In some embodiments, the inlet assembly includes a filter, shown as screen **42**, to filter particulates from flowing in the fluid passage **34**, past the screen **42**, and to the body **22** and the head portion **18**.

[0039] Referring to FIGS. **1-12**, the body **22** is configured to receive at least a portion of the inlet assembly **14** at a first end **44** of the body **22**. The body **22** is configured to receive at least a portion of the head portion **18** at a second end **46** of the body **22** to fluidly couple the inlet assembly **14** with the head portion **18**. The body **22** includes at least one housing member (e.g., casing member, wall, exterior shell, etc.), shown as housing **50**, defining an interior chamber (e.g., a space) to house one or more internal components (e.g., valves, filters, actuators, conduits, etc.) of the spray head assembly **10** therein. The housing **50** extends along a longitudinal axis A between the inlet assembly **14** and the head portion **18**.

[0040] In the embodiment of FIGS. **1-6**, the housing **50** is generally cylindrically shaped (e.g., annular). In such an embodiment, the housing **50** defines a generally circular (e.g., annular, ovular, etc.) profile (as best shown in the top view of FIG. **5** and the bottom view of FIG. **6**). In the embodiment of FIGS. **7-12**, the housing **50** is generally cuboidally (e.g., cubically) shaped having substantially planar sidewalls. In such an embodiment, the housing **50** defines a generally square (e.g., rectangular, quadrilateral, etc.) profile (as best shown in the top view of FIG. **11** and the bottom view of FIG. **12**). In yet other embodiments, the housing **50** forms a different shape or profile.

[0041] Referring to FIGS. **1-6**, the housing **50** includes a first annular portion (e.g., a shell, a casing, etc.), shown as lower shell **54**, a second annular portion (e.g., an annular member, a selector, etc.), shown as date band **58**, and a third annular portion (e.g., a shell, a casing, etc.), shown as upper shell **62**. Each of the lower shell **54**, the date band **58**, and the upper shell **62** may be rotatably coupled to each other to enable relative rotation therebetween. By way of example, the lower shell **54**, the date band **58**, and the upper shell **62** may be (i) concentric and (ii) movable (e.g., rotatable) relative to each other about the longitudinal axis A defined by the body **22**.

[0042] As shown in FIG. **13**, the lower shell **54** includes a flange **66** positioned radially inward relative to the date band **58** and at least a portion of the upper shell **62**. The flange **66** may be configured to engage with an inner radial surface of the date band **58**. In some embodiments, one or more sealing members (e.g., gaskets), shown as O-rings **70**, are coupled to the outer radial surface of the flange **66** and positioned between (e.g., radially between) the flange **66** and the date band **58**. The O-rings **70** may engage with the inner radial surface of the date band **58** and the outer radial surface of the flange **66** to form a seal (e.g., a watertight seal) therebetween. In other embodiments, the lower shell **54** is integrally formed with the upper shell **62** as a single unitary body (e.g., as a single monolithic piece that cannot be separated without damaging the lower shell **54** or the upper shell **62**) and the date band **58** is slidably engaged with the upper shell **62** and/or the lower shell **54** (e.g., in a groove and/or channel defined by the upper shell **62** and/or the lower shell **54**).

[0043] In some embodiments, the date band **58** includes a protrusion **74** along the inner radial surface thereof structured to support the one or more O-rings **70**. In some embodiments, the flange **66** is stepped radially inwards relative to an outer surface of the lower shell **54** by a step **78**. The step **78** may be structured to support the one or more O-rings **70**. The step **78** and the protrusion **74** may cooperate to support the one or more O-rings **70** thereon. A friction force between the O-rings **70**, the flange **66**, and the date band **58** may secure the orientation of the date band **58** relative to the lower shell **54** to prevent inadvertent movement of the date band **58** during use. A user may engage with the date band **58** to overcome the friction force and rotate the date band **58** relative to the lower shell **54** to a desired orientation.

[0044] Referring to FIG. **2**, the date band **58** includes the numbers “1” through “12”, shown as numbers **82** along an outer radial surface thereof. In other embodiments, the date band **58** may

include another form of service life indicator. The upper shell **62** includes an alignment feature **86** (e.g., a line, an arrow, a mark, a protrusion, etc.) along the outer radial surface thereof and located proximate the date band **58**. The numbers **82** and the alignment feature **86** may be molded, etched, engraved, embossed, debossed, adhered (e.g., as a sticker), painted, or otherwise applied to the date band **58** and the upper shell **62**, respectively. A user may rotate the date band **58** and the upper shell **62** relative to each other to substantially align one of the numbers **82** with the alignment feature **86**. In some embodiments, the numbers **82** are representative of a certain month of the year (e.g., “1” corresponding with January, “2” corresponding with February, . . . , “12” corresponding with December) such that aligning one of the numbers **82** with the alignment feature **86** is indicative of when (e.g., a day, a month, a year, etc. in which) an action should be carried out by the user. By way of example, the user may rotate the date band **58** to set a date (e.g., set a month) when one or more of the components of the spray head assembly **10** (e.g., valves, filters, actuators, conduits, etc.) should be serviced, replaced, cleaned, etc.

[0045] By way of another example, if a filter (e.g., filter assembly **144**) with a recommended life-span of six months is replaced in February, the user may align the number “8” (corresponding with August, six months from the replacement date in February) with the alignment feature **86**, thereby reminding the user to replace the filter. Additionally, or alternatively, the numbers **82** representative of the months of the year may be represented on the date band **58** in another format (e.g., spelled out, abbreviated, in braille, etc.). In some embodiments, the numbers “1” through “31” indicative of a day of the month are included on the date band **58** (e.g., instead of the numbers “1” through “12” indicative of the month) or are included on a second date band **58** of the body **22**.

[0046] In yet other embodiments, the spray head assembly **10** includes another type of indicator to visually notify a user to service the filter or another component of the spray head assembly **10**. For example, the spray head assembly **10** may include a light element (e.g., a light emitting diode, etc.) configured to indicate a remaining service life of the filter element (e.g., based on a color of the light element, etc.). In yet other embodiments, the spray head assembly **10** includes a sensor (e.g., a pressure sensor, etc.) communicably coupled to the indicator and configured to transmit a control signal to the indicator to notify the user when the filter needs to be replaced (e.g., when a pressure drop measured by the sensor exceeds or otherwise satisfies a pressure threshold, etc.).

[0047] In some embodiments, as shown in FIGS. 7-12, and as described above, the housing **50** does not include the date band **58**. In such embodiments, the housing **50** is formed as a single unitary body (e.g., shell). In other words, in such embodiments, the lower shell **54** and the upper shell **62** are integrally formed with each other as a single unitary body.

[0048] Referring to FIGS. 13 and 14, the housing **50** includes a connector flange **88** (e.g., a threaded flange, a support member, etc.) extending from a portion of the housing **50** proximate the first end **44** (e.g., from the upper shell **62**, from a portion of the housing **50** proximate the opening **108**). The connector flange **88** may extend within the interior chamber of the housing **50** (e.g., interior chamber **106**, which may also be referred to as an interior cavity, a hollow portion, etc.) in a direction towards the second end **46**. In some embodiments, the connector flange **88** may be concentric with the longitudinal axis A. The connector flange **88** includes internal threads that screw to a portion of a filter housing (e.g., filter housing **124**) to facilitate selectively coupling the filter housing with the housing **50**.

[0049] As shown in FIGS. 1-12, the head portion **18** includes a spray face **90** fluidly coupled with the body **22** and the inlet assembly **14**. The spray face **90** includes a plurality of apertures through which a plurality of nozzles **92** extend. The plurality of nozzles **92** are variously positioned about the spray face **90** and are configured to direct the fluid into a spray pattern (e.g., a spray pattern based on the arrangement of the nozzles **92**), thereby discharging the fluid from the head portion **18** and the spray head assembly **10**.

[0050] In some embodiments, as shown in FIGS. 1 and 6, the nozzles **92** are arranged in an annular pattern concentric about the longitudinal axis A. In such embodiments, the spray face **90** is

substantially circular (e.g., annular, ovalar, etc.). In other embodiments, as shown in FIGS. 7 and 12 the nozzles 92 are arranged in a prised pattern about the spray face 90. In such embodiments, the spray face 90 is substantially square (e.g., rectangular, quadrilateral, etc.). In yet other embodiments, the nozzles 92 may be arranged in a different pattern about the spray face 90.

[0051] In some embodiments, the spray face 90 includes a first set of nozzles 92 configured to discharge the fluid in a first pattern and a second set of nozzles 92 configured to discharge the fluid in a second pattern that is different from the first pattern. In such an embodiment, the fluid may be selectively discharged through the first set of nozzles 92 and/or the second set of nozzles 92.

[0052] As shown in FIGS. 13 and 14, the nozzles 92 are fluidly coupled with one or more fluid channels 96 (e.g., fluid conduits, fluid paths, etc.) configured to supply fluid received from the inlet assembly 14 and through the body 22 to the nozzles 92. In some embodiments, the channels 96 are spaced from each other such that a first channel 96 supplies fluid to first nozzles 92 (e.g., a first sub-set of nozzles 92) and a second channel 96 (e.g., a next radial outward channel 96) supplies fluid to second nozzles 92 (e.g., a second sub-set of nozzles 92) arranged radially outwards relative to the first nozzles 92. As shown, the spray face 90 includes a fastener assembly 98 (e.g., a nut and bolt assembly, a coupling assembly, a fastener mechanism, etc.) configured to couple a structure (e.g., the control member 166) including the channels 96 to an interior facing surface of the spray face 90 (e.g., a surface of the spray face 90 facing the interior chamber 106).

[0053] In the embodiment of FIGS. 13 and 14, the spray face 90 includes a connector flange 100 (e.g., threaded flange, support member, etc.) extending from the spray face 90 (e.g., from an interior facing surface of the spray face 90). The connector flange 100 may extend within the interior chamber of the housing 50 (e.g., interior chamber 106) in a direction towards the first end 44. The connector flange 100 may be positioned proximate an outer radial edge of the spray face 90. The connector flange 100 may be concentric with the longitudinal axis A. The connector flange 100 includes internal threads that screw to a portion of a filter housing (e.g., filter housing 124) to facilitate selectively coupling the filter housing with the spray face 90 and the head portion 18.

[0054] The head portion 18 includes a head shell 102 concentric with the longitudinal axis A and positioned proximate the second end 46 of the spray head assembly 10. An outer radial surface of the head shell 102 may substantially align and be substantially flush with an outer radial surface of the housing 50 (e.g., with the lower shell 54 of the housing 50) when the head portion 18 is engaged with and rotatably coupled to the body 22, such that the spray head assembly 10 defines a continuous outer profile.

[0055] The body 22 may be selectively and removably coupled with the head portion 18 such that the one or more internal components housed within the interior chamber of the housing 50 are accessible to a user without damaging the spray head assembly 10. In some embodiments, when the body 22 and the head portion 18 are decoupled (e.g., separated, removed) from each other, the inlet assembly 14 remains rotatably coupled with the body 22. Decoupling the body 22 and the head portion 18 from each other may facilitate access to the interior chamber to service, replace, clean, etc. the one or more internal components of the spray head assembly 10.

[0056] In some embodiments, the lower shell 54 of the housing 50 and the head shell 102 of the head portion 18 are detachably and rotatably coupled to one another to facilitate access to the interior chamber 106. In some embodiments, as shown in FIG. 2, the lower shell 54 of the housing 50 and the head shell 102 of the head portion 18 each include an alignment feature 103 (e.g., a line, an arrow, a mark, a protrusion, etc.) along the outer radial surface thereof. The alignment features 103 may be molded, etched, engraved, embossed, debossed, adhered (e.g., as a sticker), painted, or otherwise applied to the lower shell 54 and the head shell 102, respectively. A user may rotate the body 22 and the head portion 18 relative to each other to an alignment position to align the respective alignment features 103 of the lower shell 54 and the head shell 102.

[0057] In some embodiments, an alignment of the respective alignment features 103 of the lower shell 54 and the head shell 102 is indicative of the spray head assembly 10 being in an unlocked

configuration such that the body **22** and the head portion **18** can be freely decoupled from each other. In other embodiments, an alignment of the respective alignment features **103** of the lower shell **54** and the head shell **102** is indicative of the spray head assembly **10** being in a locked configuration such that the body **22** and the head portion **18** are coupled together.

[0058] In some embodiments, such as the embodiments shown in FIGS. 7-12, (e.g., when the housing **50** is formed as a single unitary body, the lower shell **54** and the upper shell **62** are integrally formed with each other as a single unitary body, etc.), the housing **50** and the head shell **102** do not include the alignment features **103**. In such an embodiment, an alignment of the shape (e.g., profile) of the housing **50** with the shape (e.g., profile) of the head shell **102** (e.g., rotating the housing **50** and the head shell **102** to the alignment position) is indicative of the spray head assembly **10** being in the unlocked configuration or the locked configuration. By way of example, as discussed above, the housing **50** and the head shell **102** may define generally square profiles. In such an example, aligning a first cross-sectional (e.g., square) profile of the housing **50** with a second cross-sectional (e.g., square) profile of the head shell **102** may be indicative of the spray head assembly **10** being in the unlocked configuration or the locked configuration.

[0059] As shown in FIGS. 15A-15B, and FIGS. 16-18, the body **22** and the head portion **18** may rotate relative to each other to selectively couple with each other. By way of example, any one of the head portion **18**, the body **22** (e.g., the housing **50**, the lower shell **54**, the date band **58**, or the upper shell **62**), or the inlet assembly **14** can rotate relative to any other one of the head portion **18**, the body **22**, or the inlet assembly **14**. The spray head assembly **10** includes a locking system **104** (e.g., a turn mechanism, a guide, etc.) configured to inhibit rotation of the body **22** and the head portion **18** relative to each other and selectively couple the body **22** with the head portion **18**. The locking system **104** facilitates transitioning the spray head assembly **10** between the locked configuration and the unlocked configuration.

[0060] In some embodiments, when the spray head assembly **10** is in the locked configuration such that the body **22** and the head portion **18** are coupled with each other, the body **22** and the head portion **18** may be rotated a certain angle, shown as angle θ , relative to each other to transition the spray head assembly **10** to the unlocked configuration. The angle θ may be about 30 degrees (e.g., 27 degrees, 33 degrees, etc.). In some embodiments, the angle θ is greater than or less than 30 degrees (e.g., 10 degrees, 45 degrees, 90 degrees, 180 degrees, 360 degrees, etc.). The locking system **104** may prevent the body **22** and the head portion **18** from rotating relative to each other more than the angle θ .

[0061] Referring back to FIGS. 13 and 14, the housing **50**, the head shell **102**, and the spray face **90** (e.g., interior facing surfaces of the housing **50**, the head shell **102**, and the spray face **90**) collectively define an interior chamber **106** of the spray head assembly **10**. The housing **50** includes an aperture (e.g., opening, hole, space, etc.), shown as opening **108**, proximate the first end **44** and structured to receive the ball joint **30** of the inlet assembly **14**. The inlet assembly **14** (e.g., the connector **26**) may extend through the opening **108**. In other words, when the inlet assembly **14** fluidly couples the spray head assembly **10** to the fluid delivery component, at least a portion of the inlet assembly **14** (the ball joint **30**) is positioned within (e.g., extends inside of) the interior chamber **106**, and at least a portion of the inlet assembly **14** (the connector **26**) is positioned outside of (e.g., extends through the opening **108**) the interior chamber **106**. The opening **108** is sized such that the housing **50** (and the body **22** and the head portion **18**) is pivotable about the ball joint **30** (e.g., pivotable about a center point of the ball joint **30**) relative to the inlet assembly **14** and the fluid delivery component.

[0062] In at least one embodiment, the spray head assembly **10** includes a first compressible member **112** and a second compressible member **116** positioned within the opening **108** of the housing **50**. The first compressible member **112** may be disposed along and contact an upper portion of the ball joint **30** (e.g., a portion of the ball joint **30** closer to the first end **44**). The first compressible member **112** may engage with a lip **120** of the housing **50**. The lip **120** may be

structured to inhibit translation of the first compressible member **112** within the opening **108** along the longitudinal axis A (e.g., to inhibit translation of the first compressible member **112** in a direction towards the first end **44**, etc.).

[0063] The second compressible member **116** may be disposed along and contact a lower portion of the ball joint **30** (e.g., a portion of the ball joint **30** closer to the second end **46**). The second compressible member **116** may be (i) engaged with and (ii) biased in a direction towards the first end **44** by a filter housing (e.g., filter housing **124**). Biasing the second compressible member **116** in a direction towards the first end **44** loads the first compressible member **112** between the lip **120** of the housing **50** and the ball joint **30**. This loading induces friction, which is tailorable to maintain the spray head assembly **10** (e.g., the head portion **18**) in any moved position by a user. The first and second compressible members **112**, **116** can be made of or include a resilient/compressible material, such as an elastomer or other suitable material.

[0064] Referring to FIGS. **13**, **14**, **19A**, and **19B**, the interior chamber **106** may be sized to receive a filter housing **124** therein. The filter housing **124** includes a first filter housing member (e.g., sidewall, casing, surface, connector, etc.), shown as top wall **128**, configured to selectively couple the filter housing **124** to the housing **50** within the interior chamber **106**; a second filter housing member (e.g., sidewall, casing, etc.), shown as upper sidewall **132**; and a third filter housing member (e.g., sidewall, casing, etc.), shown as lower sidewall **136**. Collectively, the top wall **128**, the upper sidewall **132**, and the lower sidewall **136** form an interior chamber (e.g., a space, a volume, etc.), shown as filter chamber **140**. The filter chamber **140** is sized to receive a filter assembly **144** (discussed below) therein. The filter assembly **144** is configured to filter the fluid supplied from the inlet assembly **14** to the head portion **18**. In some embodiments, the top wall **128**, the upper sidewall **132**, and/or the lower sidewall **136** are made from two or more sections of material. In some embodiments, the top wall **128** and the upper sidewall **132** are integrally formed with each other as a single unitary body.

[0065] As shown in FIGS. **13** and **14**, the top wall **128** includes a connector flange **148** (e.g., threaded flange, support member, etc.) extending from the top wall **128** (e.g., from an exterior facing surface of the top wall **128**). The connector flange **148** may extend within the interior chamber **106** in a direction towards the first end **44** (e.g., when the filter housing **124** is coupled to the housing **50**). The connector flange **148** may be concentric with the longitudinal axis A when the filter housing **124** is coupled to the housing **50**. The connector flange **148** includes external threads that screw to the interior threads of the connector flange **88** of the housing **50** to facilitate selectively coupling the filter housing **124** with the housing **50** and the body **22**.

[0066] The top wall **128** may include a step portion (e.g., a support surface), shown as step **152**, structured to engage with the second compressible member **116**. The step **152** may contact a bottom surface of the second compressible member **116** to support the second compressible member **116**. The step **152** may bias the second compressible member **116** in a direction towards the first end **44** when the filter housing **124** is engaged with (e.g., screwed into) the housing **50**. By way of example, the internal threads of the connector flange **88** may mesh or otherwise engage with the external threads of the connector flange **148** to couple the filter housing **124** with the housing **50**, thereby biasing the second compressible member **116** in a direction towards the first end **44**.

Biasing the second compressible member **116** in a direction towards the first end **44** against the ball joint **30** loads the first compressible member **112** between the lip **120** of the housing **50** and the ball joint **30**. This loading induces friction, which is adjustable to maintain the spray head assembly **10** (e.g., the head portion **18**) in any moved position by a user. In some embodiments, an interior facing radial surface of the connector flange **148** (e.g., a surface of the connector flange **148** opposite the external threads thereof) may engage with an exterior facing radial surface of the second compressible member **116** to support the second compressible member **116**. The top wall **128** includes an opening to receive the fluid from the fluid passage **34** of the inlet assembly **14** and supply the fluid to the filter chamber **140** and the filter assembly **144**.

[0067] The upper sidewall **132** of the filter housing **124** extends from the top wall **128**. The upper sidewall **132** may engage with an upper surface (e.g., the exterior facing surface) of the top wall **128** and an exterior facing radial surface of the connector flange **88** of the housing **50**. In some embodiments, an O-ring **154** is positioned between the top wall **128** and the upper sidewall **132**. In some embodiments, when the body **22** and the head portion **18** are coupled together, the upper sidewall **132** extends in a direction from the top wall **128** to the head shell **102**. In such an embodiment, the upper sidewall **132** overlaps at least a portion of the lower sidewall **136**. By way of example, at least a portion of the lower sidewall **136** may be received by the upper sidewall **132** (e.g., within a portion of the filter chamber **140** defined by the upper sidewall **132**) such that at least a portion of the lower sidewall **136** is positioned radially inwards relative to the upper sidewall **132**. [0068] As shown in FIGS. **13** and **14**, the lower sidewall **136** includes a shoulder **158** (e.g., a lip, a support, a flange, etc.), extending radially inward. The shoulder **158** is configured to engage with the filter assembly **144** (e.g., a bottom surface/wall of the filter assembly **144**) to support the filter assembly **144** when it is received within the filter chamber **140** of the filter housing **124**. The shoulder **158** includes an opening such that fluid filtered by the filter assembly **144** is supplied to the channels **96** and discharged through the nozzles **92**. In some embodiments, the opening of the shoulder **158** is concentric with the longitudinal axis A.

[0069] In some embodiments, the lower sidewall **136** includes a bottom flange **162** configured to engage with a control member **166**. The control member **166** may include a plurality of slots **170** variously (e.g., annularly) positioned about the control member **166**. The slots **170** may be fluidly coupled with the channels **96** and the opening defined by the shoulder **158** of the lower sidewall **136**. The slots **170** may be variously sized to control the flow rate of the fluid supplied to the channels **96** and the nozzles **92**. In other embodiments, as shown in FIG. **14**, the spray head assembly **10** does not include the control member **166**. In such embodiments, the channels **96** and the nozzles **92** are fluidly coupled directly with the opening defined by the shoulder **158**.

[0070] In some embodiments, as shown in FIG. **13**, the filter housing **124** includes a connector **174** positioned between the bottom flange **162** and the spray face **90**. The connector **174** may include exterior threads configured to engage with the connector flange **100** of the spray face **90** to selectively couple the spray face **90** (and the head portion **18**) with the filter housing **124**. In some embodiments, O-rings are positioned between (i) the connector **174** and the lower sidewall **136** and (ii) between the connector **174** and the control member **166**. In other embodiments, as shown in FIG. **14**, the connector **174** does not include exterior threads. In such an embodiment, the spray face **90** may couple to the connector **174** by way of one or more clips **178**.

[0071] As shown in FIG. **19A**, the connector **174** includes one or more fastener receivers **182** extending in a direction towards the first end **44** from the connector **174**. The fastener receivers **182** may be received by apertures **186** annularly positioned along the bottom flange **162** of the lower sidewall **136**. The fastener receivers **182** may receive a fastener **190** to couple the connector **174** with the lower sidewall **136**, thereby coupling the head portion **18** with the filter housing **124**.

[0072] Referring to FIGS. **19A** and **19B**, the spray head assembly **10** supports a common filter assembly design (e.g., a common filter housing **124**) that may be used with various different structures and/or aesthetics of the body **22** (e.g., the housing **50**) and/or the head portion **18** (e.g., the head shell **102**). By way of example, the filter housing **124** may be received in either the housing **50** of the embodiment of FIGS. **1-6** (e.g., the cylindrically shaped housing **50**) or the housing **50** of the embodiment of FIGS. **7-12** (e.g., the cuboidally shaped housing **50**). In other words, the spray head assembly **10** may be configurable both as (i) a first spray head assembly **10** including the housing **50** and the head portion **18** defining a first shape configured to receive the filter housing **124** and (ii) a second spray head assembly **10** including the housing **50** and the head portion **18** defining a second shape different than the first shape configured to receive the same filter housing **124**.

[0073] Referring to FIGS. **15B**, **20**, and **21**, when the body **22** is separated from the head portion

18, the top wall **128** and the upper sidewall **132** of the filter housing **124** may remain coupled with the body **22** and the lower sidewall **136** of the filter housing **124** may remain coupled with the head portion **18**. As shown, the locking system **104** includes a plurality of protrusions, shown as engagement features **194**. In some embodiments, the engagement features **194** extend from an interior surface of the upper sidewall **132** along the inner circumference thereof. In other embodiments, the engagement features **194** extend from an exterior surface of the lower sidewall **136** along an outer circumference thereof. The locking system **104** includes engagement track (e.g., guide, slot, etc.), shown as track **198**. In some embodiments, the track **198** is positioned at an exterior surface of the lower sidewall **136** along an outer circumference thereof. In other embodiments, the track **198** is positioned at an interior surface of the upper sidewall **132** along the inner circumference thereof.

[0074] The track **198** includes a slot **202** configured to receive the engagement feature **194** and a guide **206** configured to inhibit longitudinal translation of the engagement feature **194** (e.g., translation of the body **22** relative to the head portion **18** along the longitudinal axis A) when the engagement feature **194** is received therein. The guide **206** may support the engagement feature **194** and permit rotation of the engagement feature **194** within the guide **206**, thereby permitting relative rotation of the body **22** and the head portion **18**. The guide **206** may inhibit rotation of the engagement feature **194** within the guide **206** beyond the angle θ , thereby inhibiting relative rotation of the body **22** and the head portion **18**. By way of example, the engagement feature **194** may be received within the slot **202** and the body **22** may be rotated, thereby translating the engagement feature **194** within the guide **206** and transitioning the spray head assembly **10** to the locked configuration. Similarly, the body **22** may be rotated in an opposite direction, thereby translating the engagement feature **194** within the guide **206** and transitioning the spray head assembly **10** to the unlocked configuration. In the unlocked configuration, the engagement feature **194** and the slot **202** may be aligned such that the body **22** and the head portion **18** can be separated from each other. In some embodiments, the locking system **104** includes detents, notches, or similar features to provide tactile and audible feedback to the user when the spray head assembly **10** is transitioned between the locked configuration and the unlocked configuration. By way of example, when the body **22** and the head portion **18** are secured together, one or more components or features of the locking system **104** produce a tactile and/or audible feedback perceptible by the user. The tactile and/or audible feedback may be indicative of the spray head assembly **10** being transitioned to the locked configuration.

[0075] The filter housing **124** is configured to receive the filter assembly **144** within the filter chamber **140**. When the body **22** is separated from the head portion **18**, the upper sidewall **132** and the lower sidewall **136**, at least partially defining the filter chamber **140**, are separated such that the filter chamber **140** is accessible. The filter assembly **144** may be placed within the filter chamber **140**. The body **22** and the head portion **18** may be coupled together with the filter assembly **144** received within the filter chamber **140** to transition the spray head assembly **10** to the locked configuration, thereby securing the filter assembly **144** within the filter chamber **140**. In some embodiments, one or more O-rings **208** positioned between the exterior surface of the filter assembly **144** and the interior surface of the filter housing **124** to create a water-tight seal therebetween.

[0076] As shown in FIGS. **13**, **14**, and **20-24**, the filter assembly **144** includes a fluid passage **210** configured to receive the fluid from the inlet assembly **14** and supply the fluid to the channels **96** to be discharged through the nozzles **92**. The filter assembly **144** is configured to filter particulates, impurities, contaminants, and particles from the fluid as it is supplied therethrough. The filter assembly **144** includes a first filter (e.g., mesh layer), shown as first screen **214**, positioned proximate the inlet assembly **14** (e.g., when the filter assembly **144** is received within the filter housing **124**) and a second filter (e.g., mesh layer), shown as second screen **218**, positioned proximate the spray face **90** (e.g., when the filter assembly **144** is received within the filter housing

124). The first and second screens **214**, **218** are configured to prevent large particulates such as dirt, sand, stones, or any other debris from flowing in the fluid passage **210**, past the first and second screens **214**, **218**, and to the nozzles **92**. In some embodiments, the second screen **218** is finer than the first screen **214** to prevent smaller particulates from flowing in the fluid passage to the nozzles **92**. The filter assembly **144** may be positioned within the filter chamber **140** such that the fluid filtered by the filter assembly **144** is supplied directly to the channels **96** to be discharged through the nozzles **92** (e.g., without passing or flowing through an intervening fluid passage defined by the body **22** or a separate conduit).

[0077] Referring to FIGS. **20-23**, the spray head assembly **10** facilitates use of (e.g., installation of) the filter assembly **144** within the filter housing **124** the spray head assembly **10** with various different body **22** and/or head portion **18** structures and/or aesthetics. As shown in FIGS. **20** and **21**, a common filter assembly **144** is received within the filter housing **124** when the housing **50** is generally cylindrically shaped (e.g., such as in the embodiment of FIGS. **1-6**). As shown in FIG. **22**, the common filter assembly **144** may be received within the filter housing **124** when the spray head assembly **10** includes a structure **220** that is generally cuboidally (e.g., cubically) shaped having substantially planar sidewalls (e.g., such as in the embodiment of FIGS. **7-12**)

[0078] As shown in FIG. **24**, the filter assembly **144** includes a third filter, shown as Kinetic Degradation Fluxion (KDF) layer **222**, positioned within the fluid passage **210** below the first screen **214**. The KDF layer **222** may be configured to remove or otherwise filter chlorine and organic constituents (e.g., bacteria) from the fluid. As shown, the filter assembly **144** includes a fourth filter, shown as charcoal layer **226**, positioned within the fluid passage **210** below the KDF layer **222**. The charcoal layer **226** may be configured to remove or otherwise filter chlorine and heavy metal (e.g., lead, mercury, zinc, copper, etc.) from the fluid. As shown, the filter assembly **144** includes a fifth filter, shown as antiscalant layer **230**, positioned within the fluid passage **210** below the charcoal layer **226**. The antiscalant layer **230** may be configured to reduce the formation of scale within the spray head assembly **10**. In some embodiments, the KDF layer **222**, the charcoal layer **226**, and the antiscalant layer **230** are otherwise positioned or ordered. In other embodiments, the filter assembly **144** may include other materials to facilitate contaminant removal or otherwise alter fluid properties of the fluid to improve a user's overall showering experience.

[0079] As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the application as recited in the appended claims.

[0080] It should be noted that the term “exemplary” as used herein to describe various embodiments is intended to indicate that such embodiments are possible examples, representations, and/or illustrations of possible embodiments (and such term is not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0081] The terms “coupled,” “connected,” and the like, as used herein, mean the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members or the two members and any additional intermediate members being integrally formed as a single unitary body with one another or with the two members or the two members and any additional intermediate members being attached to one another.

[0082] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below,” etc.) are merely used to describe the orientation of various elements in the figures. It should be

noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure. [0083] It is important to note that the construction and arrangement of the apparatus and control system as shown in the various exemplary embodiments is illustrative only. Although only a few embodiments have been described in detail in this disclosure, those skilled in the art who review this disclosure will readily appreciate that many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.) without materially departing from the novel teachings and advantages of the subject matter described herein. For example, elements shown as integrally formed may be constructed of multiple parts or elements, the position of elements may be reversed or otherwise varied, and the nature or number of discrete elements or positions may be altered or varied. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments.

[0084] Other substitutions, modifications, changes and omissions may also be made in the design, operating conditions and arrangement of the various exemplary embodiments without departing from the scope of the present application. For example, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein.

Claims

1. A spray head assembly comprising: an inlet assembly configured to receive a fluid; a body fluidly coupled with the inlet assembly; a head portion fluidly removably coupled with the body, the head portion including a plurality of nozzles configured to discharge the fluid from the spray head assembly, the body and the head portion together defining an interior chamber; and a filter assembly configured to filter the fluid supplied from the inlet assembly to the head portion, the filter assembly configured to be received within the interior chamber.
2. The spray head assembly of claim 1, further comprising a filter housing configured to be received in the interior chamber, the filter housing including a first sidewall and a second sidewall defining a filter chamber, the first sidewall configured to couple with the body, and the second sidewall configured to couple with the head portion.
3. The spray head assembly of claim 2, wherein the filter assembly is configured to be received within the filter chamber, and wherein, when the body and the head portion are separated, the second sidewall remains coupled with the head portion and the first sidewall remains coupled with the body, thereby facilitating access to the filter chamber to replace the filter assembly.
4. The spray head assembly of claim 2, wherein the filter housing includes a top wall configured to couple the first sidewall with the body.
5. The spray head assembly of claim 4, wherein the body includes a connector flange extending within the interior chamber and configured to couple to the top wall.
6. The spray head assembly of claim 5, wherein the first sidewall is configured to engage with an upper surface of the top wall and an exterior facing radial surface of the connector flange.
7. The spray head assembly of claim 2, wherein the inlet assembly includes a ball joint configured to permit selective movement of the body relative to the inlet assembly.
8. The spray head assembly of claim 7, further comprising a compressible member disposed between the ball joint and the filter housing and configured to induce friction between the ball joint and the filter housing to retain a position of the body relative to the inlet assembly.
9. The spray head assembly of claim 2, wherein when the body and the head portion are coupled together, the first sidewall overlaps at least a portion of the second sidewall.
10. The spray head assembly of claim 2, wherein the second sidewall includes a shoulder configured to engage with the filter assembly to support the filter assembly within the filter chamber.

- 11.** The spray head assembly of claim 1, wherein the spray head assembly is configurable both as (i) a first spray head assembly including the body and the head portion defining a first shape configured to receive the filter assembly and (ii) a second spray head assembly including the body and the head portion defining a second shape different than the first shape configured to receive the filter assembly.
 - 12.** The spray head assembly of claim 2, further comprising at least one channel disposed between the second sidewall and a spray face of the head portion, the at least one channel configured to supply the fluid to the plurality of nozzles.
 - 13.** The spray head assembly of claim 12, wherein the fluid filtered by the filter assembly is supplied directly to the at least one channel.
 - 14.** The spray head assembly of claim 1, further comprising a locking system configured to selectively couple the body with the head portion.
 - 15.** The spray head assembly of claim 14, wherein the locking system includes an engagement feature along the body or the head portion and a guide along the other one of the body or the head portion and configured to receive the engagement feature.
 - 16.** The spray head assembly of claim 1, wherein the body includes a first portion and a second portion movable relative to the first portion, and wherein the second portion includes an indicator indicative of when an action should be carried out by a user based on a position of the second portion relative to the first portion.
 - 17.** A spray head assembly comprising: an inlet assembly configured to receive a fluid; a body fluidly coupled with the inlet assembly; a head portion fluidly removably coupled with the body, the head portion including a plurality of nozzles configured to discharge the fluid from the spray head assembly, the body and the head portion defining an interior chamber; a filter housing defining a filter chamber and configured to be received within the interior chamber; and a filter assembly configured to filter the fluid supplied from the inlet assembly to the head portion, the filter assembly configured to be received within the filter chamber, wherein the spray head assembly is configurable both as (i) a first spray head assembly including the body and the head portion defining a first shape configured to receive the filter housing and (ii) a second spray head assembly including the body and the head portion defining a second shape configured to receive the filter housing, and wherein the second shape is different than the first shape.
 - 18.** The spray head assembly of claim 17, wherein, when the body and the head portion are separated, a first portion of the filter housing remains coupled with the head portion and a second portion of the filter housing remains coupled with the body, thereby facilitating access to the filter chamber to replace the filter assembly.
 - 19.** A spray head assembly comprising: an inlet assembly configured to receive a fluid; a body fluidly coupled with the inlet assembly, the body including a connector flange; a head portion fluidly coupled with the body, the head portion including a plurality of nozzles configured to discharge the fluid from the spray head assembly; a filter housing defining a filter chamber, the filter housing including: a first sidewall; a top wall configured to couple to the connector flange to couple the first sidewall with the body; and a second sidewall configured to couple to the head portion and overlap with at least a portion of the first sidewall; and a filter assembly configured to filter the fluid supplied from the inlet assembly to the head portion, the filter assembly configured to be received within the filter chamber of the filter housing.
 - 20.** The spray head assembly of claim 19, wherein the body and the head portion are configured to selectively couple with each other, and wherein, when the body and the head portion are separated, the second sidewall remains coupled with the head portion and the first sidewall and the top wall remain coupled with the body, thereby facilitating access to the filter chamber to replace the filter assembly.
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